The Invariance Principle — problem 1.54

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Initially, we have three piles with a, b, and c chips, respectively. In one step, you may transfer one chip from any pile with x chips onto any other pile with y chips. Let $d = y - x + 1$. If $d > 0$, the bank pays you d dollars. If $d < 0$, you pay the bank $|d|$ dollars. Repeating this step several times you observe that the original distribution of chips has been restored. What maximum amount can you have gained at this stage ?